

Department Overview

- **Department Name:** Department of Microbiology
- **Faculty:** Faculty of Life Sciences & Informatics
- **Core Academic Framework:** The degree programs focus on training students in essential microbiological techniques while highlighting applied areas of the subject and their distinct relationship with other life sciences disciplines.
- **Professional Significance:** Microbiology is essential for medical health sciences, agricultural advancement, environmental management, marine sciences, and the rapidly scaling biotechnology industry.
- **Career Pathways:** Graduates are heavily demanded to fill positions such as research scientists, clinical laboratory professionals, industrial quality control microbiologists, public health officers, bio-medical equipment technical consultants, and academicians across public and private sector universities.
- **Target Industries:** Pharmaceuticals, food & beverage processing plants, vaccine development setups, healthcare product creation, agricultural disease prevention, and bioremediation/renewable energy facilities.

1. Bachelor of Science in Microbiology (BS Microbiology)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Medical) with Chemistry & Physics from any recognized board (or an equivalent qualification) with at least **45% marks**.

Degree Requirements

- **Total Credit Hours:** 134
- **Total Courses Required:** 42
- **Minimum Academic Standing:** CGPA ≥ 2.0

Curriculum Plan (Program Schema by Semester)

Note: Credit hour listings follow the format Total Credits (Theory + Lab).

- **1st Semester:** Functional English 3(3+0), Pakistan Studies 2(2+0), Mathematics- I (Algebra) 3(3+0), Biochemistry-I 3(2+1), Human Physiology-I 3(2+1), Fundamentals of Microbiology-I 4(3+1). **Total Credit Hours: 18**
- **2nd Semester:** Communication Skills 3(3+0), Islamic Studies / Ethics 2(2+0), Mathematics-II (Calculus) 3(3+0), Biochemistry-II 3(2+1), Human Physiology-II 3(2+1), Fundamentals of Microbiology-II 4(3+1). **Total Credit Hours: 18**
- **3rd Semester:** Introduction to Computer 3(2+1), Technical Writing and Presentation Skills 3(3+0), Genetics 3(2+1), Ecology and Ecosystem 3(2+1), Microbial Taxonomy 3(2+1), Fundamentals of Biotechnology 3(2+1). **Total Credit Hours: 18**

- **4th Semester:** Biodiversity of Animals 3(2+1), Current Environmental Issues 3(3+0), General Virology 3(2+1), Cell Biology 3(2+1), Mycology 3(2+1). **Total Credit Hours: 15**
- **5th Semester:** Biosafety and Risk Management 3(3+0), Immunology 3(2+1), Microbial Anatomy & Physiology 3(2+1), Epidemiology 3(2+1), Molecular Biology 3(2+1). **Total Credit Hours: 15**
- **6th Semester:** Research Methodology 3(3+0), Soil Microbiology 4(3+1), Food Microbiology 3(2+1), Fresh Water Microbiology 3(2+1), Clinical Bacteriology 4(3+1). **Total Credit Hours: 17**
- **7th Semester:** Molecular Mechanism of Antimicrobial Drugs 4(3+1), Marine Microbiology 3(2+1), Genetics Engineering 4(3+1), Medical Microbiology 3(2+1), Applied Immunology 3(2+1). **Total Credit Hours: 17**
- **8th Semester:** Bacterial Genetics 4(3+1), Research Project / Internship 6(0+6), Biostatistics 3(2+1), Plant Pathology 3(2+1). **Total Credit Hours: 16**

2. Master of Science in Microbiology (MS Microbiology)

Admission Requirements

- **Academic Background:** BS Microbiology or an equivalent undergraduate degree with a minimum graduation CGPA ≥ 2.0 , OR a minimum of 16 years of education in a relevant Life Sciences discipline passed with at least **50% marks**.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Coursework Requirements:** 8 advanced postgraduate courses.
- **Minimum Academic Standing:** CGPA ≥ 2.5